

Supplementary Material for A Masked Mixture Model for Compact and Accurate Matrix Factorization

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Abstract

This supplementary material provides additional theoretical details and experimental settings for the paper “A Masked Mixture Model for Compact and Accurate Matrix Factorization.”

A Verifying Linear Independence

This section verifies condition (C2) in Theorem 1 of the main paper for the mask family

$$f_k(r; s) = \frac{1}{K} f\left(\frac{k}{K}(r - s)\right), \quad k = 1, \dots, K,$$

where $f : \mathbb{R} \rightarrow \mathbb{R}$ is a common smooth base function. The prefactor $1/K$ is immaterial for linear independence and will be ignored in the proofs.

Notation. Fix any $r \in [R]$. Define $t := r - s$. Then the family $\{s \mapsto f_k(r; s)\}_{k=1}^K$ is linearly independent on an interval in s if and only if the family $\{t \mapsto f(\alpha_k t)\}_{k=1}^K$ is linearly independent on the corresponding interval in t , where $\alpha_k := k/K$.

LEMMA 1 (AFFINE CHANGE OF VARIABLES PRESERVES LINEAR INDEPENDENCE). *Let $\{g_k\}_{k=1}^K$ be functions on an interval $\mathcal{I} \subset \mathbb{R}$, and let $\phi(t) = at + b$ with $a \neq 0$. If $\{g_k\}$ are linearly independent on $\phi(\mathcal{I})$, then $\{g_k \circ \phi\}$ are linearly independent on $\mathcal{I}$.*

PROOF. If $\sum_{k=1}^K c_k (g_k \circ \phi)(t) = 0$ for all $t \in \mathcal{I}$, then for all $u \in \phi(\mathcal{I})$, $\sum_{k=1}^K c_k g_k(u) = 0$ by setting $u = \phi(t)$. Linear independence of $\{g_k\}$ implies all $c_k = 0$. $\square$

Thus, it suffices to prove linear independence of $\{t \mapsto f(\alpha_k t)\}_{k=1}^K$ for $\alpha_k = k/K$.

A.1 Trigonometric Bases: $f(t) = \sin(t), \cos(t)$

LEMMA 2 (DISTINCT EXPONENTIALS ARE LINEARLY INDEPENDENT). *Let $\lambda_1, \dots, \lambda_M \in \mathbb{C}$ be pairwise distinct. Then $\{t \mapsto e^{\lambda_m t}\}_{m=1}^M$ are linearly independent over $\mathbb{C}$ on any open interval.*

PROOF. Suppose $\sum_{m=1}^M a_m e^{\lambda_m t} = 0$ for all t in an open interval. By analyticity, the identity holds for all $t \in \mathbb{R}$. Evaluating derivatives at $t = 0$ gives for each $\ell = 0, 1, \dots, M-1$,

$$0 = \frac{d^\ell}{dt^\ell} \sum_{m=1}^M a_m e^{\lambda_m t} \Big|_{t=0} = \sum_{m=1}^M a_m \lambda_m^\ell.$$

This is a Vandermonde system $V\mathbf{a} = \mathbf{0}$ with $V_{\ell m} = \lambda_m^\ell$. Since λ_m are distinct, $\det(V) = \prod_{1 \leq p < q \leq M} (\lambda_q - \lambda_p) \neq 0$, hence $a_m = 0$ for all m . $\square$

PROPOSITION 3 (LINEAR INDEPENDENCE FOR SINUSOIDAL BASES). *Let $\alpha_1, \dots, \alpha_K > 0$ be pairwise distinct. Then $\{t \mapsto \sin(\alpha_k t)\}_{k=1}^K$ and $\{t \mapsto \cos(\alpha_k t)\}_{k=1}^K$ are linearly independent on any open interval. In particular, this holds for $\alpha_k = k/K$.*

PROOF. We prove the sine case; the cosine case is analogous. Assume $\sum_{k=1}^K c_k \sin(\alpha_k t) = 0$ for all t in an open interval. By analyticity, the identity holds for all $t \in \mathbb{R}$. Using $\sin(\alpha t) = \frac{e^{i\alpha t} - e^{-i\alpha t}}{2i}$, we obtain

$$0 = \sum_{k=1}^K c_k \sin(\alpha_k t) = \sum_{k=1}^K \frac{c_k}{2i} e^{i\alpha_k t} + \sum_{k=1}^K \left(-\frac{c_k}{2i}\right) e^{-i\alpha_k t}.$$

The exponents $\{i\alpha_k\}_{k=1}^K \cup \{-i\alpha_k\}_{k=1}^K$ are pairwise distinct because $\alpha_k > 0$ are distinct. By Lemma 2, all coefficients must be zero, hence $c_k = 0$ for all k . $\square$

Taking $f(t) = \sin(t)$ or $f(t) = \cos(t)$ and $\alpha_k = k/K$, Proposition 3 and Lemma 1 imply that $\{s \mapsto f_k(r; s)\}_{k=1}^K$ are linearly independent for every fixed r .

A.2 Gaussian Base: $f(t) = \exp(-t^2/(2\sigma^2))$

PROPOSITION 4 (LINEAR INDEPENDENCE FOR GAUSSIAN BASE). *Fix $\sigma > 0$ and let $\beta_1, \dots, \beta_K > 0$ be pairwise distinct. Then the functions $\{t \mapsto \exp(-\beta_k t^2/(2\sigma^2))\}_{k=1}^K$ are linearly independent on any open interval. In particular, this holds for $\beta_k = \alpha_k^2$ with $\alpha_k = k/K$.*

PROOF. Assume $\sum_{k=1}^K c_k \exp(-\beta_k t^2/(2\sigma^2)) = 0$ for all t in an open interval, hence for all $t \in \mathbb{R}$ by analyticity. Expand at $t = 0$:

$$\exp\left(-\frac{\beta_k}{2\sigma^2} t^2\right) = \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{\beta_k}{2\sigma^2}\right)^n t^{2n}.$$

Equating the coefficient of t^{2n} for $n = 0, 1, \dots, K-1$ yields

$$\sum_{k=1}^K c_k \left(\frac{\beta_k}{2\sigma^2}\right)^n = 0, \quad n = 0, 1, \dots, K-1.$$

This is a Vandermonde system in the distinct scalars $\{\beta_k/(2\sigma^2)\}_{k=1}^K$, hence the only solution is $c_k = 0$ for all k . $\square$

With $f(t) = \exp(-t^2/(2\sigma^2))$ and $\alpha_k = k/K$, we have $f(\alpha_k t) = \exp(-\alpha_k^2 t^2/(2\sigma^2))$ with distinct α_k^2 , so Proposition 4 and Lemma 1 imply (C2).

A.3 Logistic Sigmoid Base: $f(t) = \sigma(t) = \frac{1}{1+e^{-t}}$

PROPOSITION 5 (LINEAR INDEPENDENCE FOR LOGISTIC SIGMOID BASE). *Let $f(t) = \sigma(t) = \frac{1}{1+e^{-t}}$. Then $\{t \mapsto \sigma(kt)\}_{k=1}^K$ are linearly independent on any open interval. Consequently, $\{t \mapsto \sigma(\alpha_k t)\}_{k=1}^K$ are linearly independent for $\alpha_k = k/K$, and hence $\{s \mapsto f_k(r; s)\}_{k=1}^K$ are linearly independent for every fixed r .*

PROOF. Suppose $\sum_{k=1}^K c_k \sigma(kt) = 0$ for all t in an open interval, hence for all $t \in \mathbb{R}$ by analyticity. Let $w := e^{-t} \in (0, \infty)$. Then $\sigma(kt) = \frac{1}{1+e^{-kt}} = \frac{1}{1+w^k}$, so

$$\sum_{k=1}^K \frac{c_k}{1+w^k} = 0 \quad \text{for all } w > 0.$$

Multiply both sides by $Q(w) := \prod_{j=1}^K (1+w^j)$ to obtain a polynomial identity

$$\sum_{k=1}^K c_k \prod_{\substack{j=1 \\ j \neq k}}^K (1+w^j) = 0 \quad \text{for all } w > 0.$$

Since the left-hand side is a polynomial of w that vanishes on an infinite set, it is identically zero as a polynomial. Define $P_k(w) := \prod_{j \neq k} (1+w^j)$. Its degree is

$$\deg(P_k) = \sum_{j \neq k} j = \frac{K(K+1)}{2} - k,$$

which is strictly decreasing in k , hence the degrees $\deg(P_k)$ are pairwise distinct. Let k^* be the smallest index with $c_{k^*} \neq 0$ (if any). Then among $\{P_k\}_{k=k^*}^K$, the polynomial P_{k^*} has the largest degree, and its leading coefficient is 1. Thus the coefficient of the highest-degree monomial in $\sum_{k=k^*}^K c_k P_k(w)$ equals c_{k^*} , which must be zero. This contradiction implies all $c_k = 0$, proving linear independence. Finally, applying Lemma 1 yields the claim for $\{s \mapsto f_k(r; s)\}$. $\square$

B Non-Degeneracy Instantiation

We provide a simple instantiation showing that the non-degeneracy premise in Theorem 2 of the main paper is mild. In particular, the injectivity of $\mathcal{L}_\theta$ on the off-diagonal subspace holds for an open (and thus generic) set of parameters for many standard real-analytic shift-mask families.

Recall the off-diagonal subspace

$$\text{off} := \{C \in \mathbb{R}^{R \times R} : \text{diag}(C) = 0\}.$$

We show that, when $R \leq I$ and $R \leq J$, there exists a parameter point θ_0 for which the restricted linear map $\mathcal{L}_{\theta_0} : \text{off} \rightarrow \mathbb{R}^{I \times J}$ is injective.

Claim. Assume $R \leq I, J$. Then there exists θ_0 such that $\mathcal{L}_{\theta_0}$ is injective on off .

Construction. Choose factors $U \in \mathbb{R}^{I \times R}$ and $V \in \mathbb{R}^{J \times R}$ with one-hot columns:

$$U(:, r) = e_r^{(I)} \in \mathbb{R}^I, \quad V(:, r) = e_r^{(J)} \in \mathbb{R}^J, \quad r \in [R],$$

where $e_r^{(\cdot)}$ denotes the r -th standard basis vector. This choice is feasible precisely because $R \leq I, J$. With this construction, the rank-one outer product $U(:, p)V(:, q)^\top$ equals the coordinate matrix $E_{p,q}$ supported only on entry (p, q) .

Fix any real-analytic, non-constant shift-mask family and consider the resulting masks $\{M_k^U\}_{k=1}^K$ and $\{M_k^V\}_{k=1}^K$. For each pair $p \neq q$, define the scalar coefficient

$$\Delta_{p,q}(\theta) := \sum_{k=1}^K (M_k^U(p, q) M_k^V(q, q) - M_k^U(p, p) M_k^V(q, p)).$$

A direct differentiation of $\widehat{X}_\theta(\exp(tC))$ at $t = 0$ (using the one-hot structure of U, V) shows that, for any $C \in \text{off}$,

$$(\mathcal{L}_\theta(C))_{p,q} = \Delta_{p,q}(\theta) C_{p,q}, \quad \text{for all } p \neq q,$$

i.e., each off-diagonal entry (p, q) of the output depends only on the corresponding coefficient $C_{p,q}$, scaled by $\Delta_{p,q}(\theta)$.

It therefore suffices to pick θ_0 such that $\Delta_{p,q}(\theta_0) \neq 0$ for all $p \neq q$. Since each $\Delta_{p,q}(\theta)$ is a real-analytic function of the mask parameters and the base mask is non-constant, $\Delta_{p,q}$ is not identically zero (for generic choices of the remaining parameters). Consequently, the zero set $\{\theta : \Delta_{p,q}(\theta) = 0\}$ has empty interior. Because there are finitely many pairs (p, q) with $p \neq q$, we can choose θ_0 outside the finite union of these zero sets, ensuring $\Delta_{p,q}(\theta_0) \neq 0$ simultaneously for all $p \neq q$.

Under this choice, if $\mathcal{L}_{\theta_0}(C) = 0$, then $\Delta_{p,q}(\theta_0) C_{p,q} = 0$ for every $p \neq q$, which forces $C_{p,q} = 0$ for all off-diagonal entries. Hence $C = 0$ in off , proving injectivity of $\mathcal{L}_{\theta_0}$ on off . $\square$

The argument relies only on (i) real-analytic dependence on the mask parameters and (ii) non-constancy of the underlying mask base. Therefore, Theorem 2 of the main paper is applicable to a wide spectrum of commonly employed real-analytic shift-mask families, including trigonometric, Gaussian, and logistic sigmoid bases.

C Experiment Settings

All experiments are conducted on a single-GPU machine equipped with an Intel Core i7-10700KF @ 3.80GHz CPU, 32GB RAM, and an NVIDIA GeForce RTX 3070 Ti GPU. Our implementation uses PyTorch 2.3.0+cu121 running on Python 3.11. We use Relative Frobenius Error for reconstruction and Root Mean Squared Error (RMSE) for matrix completion: for an input matrix X and its estimation $\widehat{X}$,

$$\text{RelErr}_F(X, \widehat{X}) = \frac{\|X - \widehat{X}\|_F}{\|X\|_F},$$

where $\|\cdot\|_F$ denotes the Frobenius norm, and

$$\text{RMSE}(\Omega_{\text{test}}) = \sqrt{\frac{1}{|\Omega_{\text{test}}|} \sum_{(i,j) \in \Omega_{\text{test}}} (X_{ij} - \widehat{X}_{ij})^2}.$$

For runtime, we measure the average per-epoch wall-clock time. All reported results are averaged over five independent runs with random seeds.

We preprocess the two real-world reconstruction datasets as follows. For YALE-B, we downsample cropped face images to 48×48 and stack vectorized images as columns. For REUTERS, we build a document-term matrix using the top 2,000 words and 3,000 documents; we treat zero entries as meaningful absences and reconstruct the full matrix.

C.1 Baselines

C.1.1 Reconstruction Baselines. We compare against the following baselines, each tuned to its best-performing hyperparameters (e.g., learning rate, regularization strength) so as to achieve strong performance comparable to truncated SVD.

- **SVD** [3, 13]: Given $X \in \mathbb{R}^{I \times J}$, let $X = U \Sigma V^\top$ be its singular value decomposition. The rank- B truncated SVD baseline is

$$\hat{X}_{\text{SVD}} = U_{(:,1:B)} \Sigma_{(1:B,1:B)} V_{(:,1:B)}^\top,$$

which is the best rank- B approximation in Frobenius norm for fully observed reconstruction.

- **MF-SGD** [8]: We learn $U \in \mathbb{R}^{I \times B}$ and $V \in \mathbb{R}^{J \times B}$ by minimizing the squared error on observed entries Ω :

$$\min_{U,V} \sum_{(i,j) \in \Omega} (X_{ij} - u_i^\top v_j)^2, \quad \hat{X}_{ij} = u_i^\top v_j,$$

optimized with stochastic gradient descent.

- **MF-Ridge** [5, 21]: We learn U, V by solving an ℓ_2 -regularized objective:

$$\min_{U,V} \sum_{(i,j) \in \Omega} (X_{ij} - u_i^\top v_j)^2 + \lambda (\|U\|_F^2 + \|V\|_F^2),$$

where λ is tuned. This baseline can be optimized via alternating ridge-regression updates, where each subproblem is a ridge regression in closed form.

- **MF-Bias** [8]: We augment MF with per-row and per-column bias vectors $a \in \mathbb{R}^I$ and $b \in \mathbb{R}^J$:

$$\hat{X}_{ij} = \mu + a_i + b_j + u_i^\top v_j,$$

where μ is a global intercept (e.g., mean of observed entries) and (a, b, U, V) are learned by minimizing the squared error on Ω (optionally with ℓ_2 regularization), using gradient-based optimization. We use $U \in \mathbb{R}^{I \times (B-1)}$ and $V \in \mathbb{R}^{J \times (B-1)}$ to account for the additional user/item bias parameters.

C.1.2 Matrix Completion Baselines. We describe the baselines used for the matrix completion task as follows.

Matrix Factorization & Probabilistic Models:

- **PMF** [10]: Probabilistic Matrix Factorization models the user-item interaction matrix using a probabilistic framework with Gaussian priors on the latent factors to handle sparse data effectively.
- **SVD++** [7]: An extension of standard matrix factorization that incorporates implicit feedback (e.g., the history of rated items) into the user representation to capture non-explicit preference signals.
- **BiasMF** [8]: A baseline method that improves factorization accuracy by explicitly modeling scalar biases for users and items to account for systematic deviations in rating tendencies.
- **LLORMA** [9]: Local Low-Rank Matrix Approximation assumes that the rating matrix is locally low-rank, and approximates it via a weighted combination of multiple local matrix factorization models restricted to specific sub-regions.

Deep Learning-based Models:

- **RBM** [12]: One of the earliest deep learning approaches, utilizing Restricted Boltzmann Machines to model the distribution of user preferences by treating ratings as ordinal values.

- **AutoRec** [14]: An autoencoder-based collaborative filtering model that takes user or item vectors as input and trains the network to reconstruct the observed ratings in the output layer.
- **NNMF** [2]: Neural Network Matrix Factorization replaces the simple dot product of latent factors with a multi-layer feed-forward neural network to learn complex non-linear interactions between users and items.
- **CF-NADE** [20]: A neural autoregressive architecture that models the joint probability of ratings, predicting values sequentially based on observed ratings to capture parameter sharing across different ratings.

Graph Neural Network-based Methods:

- **sRMGCNN** [11]: A geometric matrix completion approach that combines multi-graph convolutions over user/item graphs with a recurrent module to model a learnable diffusion process on the rating matrix, capturing local graph-structured patterns for improved prediction.
- **GC-MC** [16]: Graph Convolutional Matrix Completion formulates the task as link prediction on a bipartite graph, using a graph convolutional encoder to produce node embeddings and a bilinear decoder for rating prediction.
- **STAR-GCN** [18]: Stacked and Reconstructed GCN addresses the label leakage issue in GNN-based recommendation by employing a stack of graph encoder-decoders to learn low-dimensional embeddings from masked graphs.
- **IGMC** [19]: Inductive Graph Matrix Completion extracts enclosing subgraphs around target user-item pairs and feeds them into a GNN, allowing the model to learn structural patterns that generalize to unseen nodes.
- **IDCF-GC** [17]: An inductive collaborative filtering framework that explicitly preserves the specific features of users and items while modeling their interactions via graph convolution to support new node updates.
- **IMC-GAE** [15]: A graph autoencoder-based approach designed for inductive matrix completion, which incorporates node attribute information to predict ratings for users or items not seen during training.
- **GHRS** [1]: Graph-based Hybrid Recommendation System constructs a heterogeneous graph incorporating users, items, and their attributes, applying GNNs to capture complex semantic relationships for rating prediction.
- **MoRGH** [22]: A recent method that employs GNNs on heterogeneous graphs constructed from rich auxiliary data (e.g., movie metadata) to effectively capture high-order structural information for accurate recommendations.

For the two recent baselines, GHRS and MoRGH, we are unable to obtain official implementations or sufficient auxiliary details (e.g., preprocessing and hyperparameter configurations) to run the models under our experimental setup. Therefore, for these methods we report the RMSE as presented in the original papers. For classical baselines, when our re-implementation and execution do not reproduce the reported performance despite careful tuning, we likewise report the best results from the corresponding papers.

C.2 Hyperparameters

C.2.1 Hyperparameters for Reconstruction. For reconstruction (the reconstruction experiments in the main paper), all gradient-based methods (except MF-SGD) are optimized using Adam [6] with learning rate $\eta = 0.02$ for synthetic datasets and $\eta = 0.1$ for real-world datasets. The base factor matrices are initialized randomly, and all shift parameters of MMF are initialized to zero. Except for SVD, all methods are trained with gradient-based optimization for 2000 epochs to ensure convergence.

Mask family. For reconstruction on random dense matrices $\{S_n\}_{n=6}^{10}$, we use a mask ratio $K/B = 0.6$ and the sinusoidal mask family $f^U(x) = f^V(x) = \sin(x)$. For reconstruction on heterogeneous matrices $\{H_b\}_{b=2}^6$, we use a mask ratio $K/B = 0.4$ and the Gaussian mask family $f^U(x) = f^V(x) = \frac{\exp(-x^2/2)}{\sqrt{2\pi}}$. For the YALE-B and REUTERS datasets, we use the sinusoidal mask family and set $K/B = 0.5$. Here, B denotes the budget parameter (total parameter count $= (I + J)B$), and K is the number of mask components; in practice K is set according to the specified ratio. Although one could choose different mask families for the left and right factors (i.e., $f^U \neq f^V$), we did not observe a consistent or statistically meaningful improvement from such asymmetric choices in preliminary experiments; therefore, for simplicity and reproducibility, we set $f^U = f^V$ in all reported results.

Parameter budgets. For an input matrix $X \in \mathbb{R}^{I \times J}$, we define the total parameter budget as $(I + J)B$. For each dataset, we evaluate the following budget values B :

$$\begin{aligned} S_6 : \{5, 10, 15, 20, 25, 30\}, & \quad S_7 : \{10, 20, 30, 40, 50, 60\}, \\ S_8 : \{20, 40, 60, 80, 100, 120\}, & \quad S_9 : \{40, 80, 120, 160, 200, 240\}, \\ S_{10} : \{80, 160, 240, 320, 400, 480\}. \end{aligned}$$

For heterogeneous datasets $H_2, \dots, H_6$, we use the shared set $B \in \{80, 160, 240, 320, 400, 480\}$.

C.2.2 Hyperparameters for Matrix Completion. For matrix completion (the matrix completion experiments in the main paper), all baseline methods are configured with the optimal hyperparameters suggested in their original papers. For MMF, we train for at most 100 epochs and apply early stopping with patience 3 based on validation RMSE (we use 1% of the training set as the validation set). We optimize MMF with Adam; the base factor matrices are initialized randomly, and all shift parameters are initialized to zero. Table 1 summarizes the dataset-specific hyperparameters used for MMF, including the learning rate η , the base rank R , the number K of masks, and the mini-batch size.

Mask family. For all matrix completion datasets, we use a Gaussian mask family with a positive offset: $f^U(x) = f^V(x) = \frac{\exp(-x^2/2)}{\sqrt{2\pi}} + \frac{1}{2}$. Empirically, the additive offset $\frac{1}{2}$ tends to improve stability in extremely sparse completion. Without the offset, a Gaussian mask can become overly concentrated around its peak, assigning near-zero weights to most latent dimensions and effectively silencing a large fraction of the representation; as a result, many coordinates receive weak gradients when supervision is scarce. Adding a constant upward shift keeps the mask in a soft gating regime—still emphasizing a subset of dimensions, but avoiding near-zero scaling elsewhere—thereby improving optimization stability and reducing the chance of degenerate masking on sparse data.

Table 1: Hyperparameters for MMF on matrix completion.

Dataset	Learning rate η	Base rank R	# masks K	Batch size
FLIXSTER	0.002	240	42	64
DOUBAN	0.002	360	64	128
ML-100K	0.010	600	42	4096
ML-1M	0.005	1200	64	4096
ML-10M	0.005	960	42	8192

Bias model. We use a bias-aware MMF for all completion experiments: $\hat{X}_{ij} = \mu + a_i + b_j + \hat{R}_{ij}$, where $a \in \mathbb{R}^I$ and $b \in \mathbb{R}^J$ are per-row and per-column bias vectors, respectively, and $\hat{R}_{ij}$ denotes the interaction term produced by MMF. We fix the global intercept to $\mu = 3$, the midpoint of the standard 1–5 explicit-feedback scale, following the common protocol [4].

C.2.3 Mask-Family Ablations. In the ablation study in the main paper, we ablate the mask function family using the following definitions:

$$\begin{aligned} \text{sin: } f(x) &= \sin(x), \\ \text{Gaussian: } f(x) &= \frac{\exp(-x^2/2)}{\sqrt{2\pi}}, \\ \text{tanh: } f(x) &= \tanh(x), \\ \text{triangle: } f(x) &= \max(0, 1 - |x|), \\ \text{sigmoid: } f(x) &= \sigma(x) = \frac{1}{1 + \exp(-x)}. \end{aligned}$$

We also evaluated cosine functions, but their performance is indistinguishable from sine masks, so we report only the sine results in the paper for brevity.

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